

DengarTrack

Jejak. Pantau. Dengar.

Digital Newborn Hearing Screening Management Platform

Concept Brief

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DengarTrack is a mobile-first digital workflow platform that puts the right information in the right hands at the right time -- replacing fragmented paper records, manual Excel reporting, and unencrypted WhatsApp communication with a single secure, intelligent system purpose-built for Universal Newborn Hearing Screening (UNHS) in Malaysian public hospitals.

The Problem

Universal Newborn Hearing Screening programmes across 48 Malaysian public hospitals currently operate on disconnected, manual systems: paper result cards at the cot side, Excel spreadsheets for tracking, WhatsApp groups for inter-hospital communication, and hours of manual data aggregation every month-end. The consequences are measurable and serious:

Indicator	Current status vs Target
Loss-to-follow-up (LTFU)	29% mean across 48 hospitals -- international benchmark: $\leq 10\%$
Age at intervention	>8 months mean -- JCIH benchmark: ≤ 6 months
PDPA compliance	Active violation -- clinical data shared via consumer WhatsApp daily
Monthly MoH reporting	2-4 hours manual aggregation per coordinator per hospital per month
Data visibility	No real-time dashboard for coordinators, hospital management, or KKM

1. Use by Healthcare Workers -- Screening Personnel

DengarTrack is designed for the people doing the clinical work, not for administrators watching from above. The primary user is the ward screener -- a nurse or audiologist performing OAE or AABR screening at the cot side -- and the secondary user is the programme coordinator managing follow-up and reporting.

1A Screener -- Point-of-Care Entry

The screener's interface is built for one-handed tablet or phone use in a busy postnatal ward. Every action is completable in under 60 seconds:

- * Scan the baby's wristband QR code to auto-fill the Baby ID, ward, and date/time

- * Select the screening device used from a pre-configured dropdown
- * Tap LULUS (Pass) or RUJUK (Refer) for each ear -- large, colour-coded buttons
- * Add optional clinical notes (maximum 100 characters)
- * Submit -- data saves locally if offline, syncs automatically when connected

The screener needs no training beyond a 10-minute walkthrough. No IT literacy is assumed. The interface is fully bilingual in Bahasa Melayu and English.

1B Coordinator -- Programme Management

The coordinator sees everything the screener entered, plus the full follow-up picture across all wards:

- * Real-time follow-up queue showing all refer cases with urgency colour-coding
- * One-click action buttons: Mark Contacted, Book Appointment, Escalate, Close
- * Audit log of every action taken -- timestamped, attributed, immutable
- * Secure in-app inter-hospital handover -- replaces WhatsApp entirely
- * Monthly MoH report generated automatically from existing data

Both roles operate through the same mobile app on Android devices -- no separate login portal or desktop dependency for routine clinical use.

2. Mobile Phone Application

DengarTrack is built as an Android-first Progressive Web App (PWA), meaning it installs directly from a browser -- no App Store approval process, no IT department intervention required at the hospital level. It runs on Android 9 and above, covering an estimated 94% of devices currently in use across KKM hospital wards.

2A Device Compatibility

- * Android smartphone and tablet -- primary platform
- * iOS (iPhone/iPad) -- secondary, Phase 2 priority
- * Desktop/laptop browser -- coordinator portal and KKM reporting dashboard
- * Minimum Android 9 | Works on low-end RM 300-400 devices
- * Screen resolution: readable and usable at 720p -- standard ward tablet spec

2B Offline-First Architecture

Ward WiFi in Malaysian public hospitals is inconsistent. DengarTrack is designed to work with zero internet connectivity:

- * All screening entry and shift summary functions work fully offline
- * Data saves to local device storage (SQLite) when offline
- * Background sync occurs automatically when connectivity is restored
- * Sync status indicator visible to screener at all times -- 'Semua Tersimpan' (All Saved) or 'X rekod belum disinkron' (X records pending sync)
- * No data is ever lost due to connectivity issues

3. Integration of Manual and Written Reports

DengarTrack does not require hospitals to abandon their existing paper records on day one. It is designed for a phased transition that respects the reality of mixed paper-digital environments in public hospitals.

Bridging Paper and Digital

Three pathways exist for bringing existing manual data into DengarTrack:

- * Pathway 1 -- Real-time entry: Screener enters result directly at the cot side at the point of screening. This is the target state.
- * Pathway 2 -- Same-day batch entry: Screener completes paper cards as usual, then enters all results into DengarTrack at the end of the shift using the same simple form. Shift Summary screen shows what has been entered.
- * Pathway 3 -- Coordinator back-entry: For historical data migration, coordinators can enter past records using a date-selectable entry form. This builds the baseline dataset for the dashboard and LTFU tracking.

Over time, as the screener becomes familiar with the app, Pathway 1 naturally replaces Pathway 2. The system does not force this transition -- it enables it.

3B KKM Report Format Compliance

The MoH monthly UNHS report format is hard-coded into DengarTrack's report generator. Every field in the KKM submission template is auto-populated from screening data already in the system:

- * Jumlah Kelahiran | Jumlah Disaring | Kadar Liputan
- * Jumlah Rujukan | Kadar Rujukan | LTFU Count
- * Patuh 1-3-6 compliance percentages | Disaring Tepat Masa
- * Data completeness check flags any gaps before export
- * One-click export as Excel (.xlsx) in KKM-standard format and PDF for hospital records

If KKM updates the report format, the template is updated centrally -- all 48 hospitals receive the update automatically without any action required at the hospital level.

4. Digital Dashboards for Reporting

DengarTrack transforms data that currently lives in disconnected Excel files into live, accessible dashboards at three levels -- the ward screener, the hospital coordinator, and the national KKM view.

4A Coordinator Dashboard

- * Coverage rate -- percentage of eligible newborns screened this month, prominently displayed
- * Pass / Refer / LTFU counts -- colour-coded tiles updated in real time

- * 1-3-6 benchmark tracker -- three horizontal progress bars showing compliance with screen by 1 month, diagnose by 3 months, intervene by 6 months
- * Six-month screening trend chart -- visual line graph, no Excel pivot table required
- * Ward-by-ward breakdown table -- coverage rate and refer rate per ward
- * Outstanding follow-up alert -- count of cases requiring action with direct link to queue

4B Hospital Admin Summary View

- * Hospital-level coverage rate versus national benchmark -- donut chart comparison
- * LTFU rate versus national average with directional indicator
- * Six-month rolling trend line
- * One-click PDF summary download for JCI/MS audit or department meetings
- * Programme coordinator accountability panel -- name and last login timestamp

KKM National Dashboard (Phase 2)

- * Aggregate national view across all participating hospitals
- * State-level and hospital-level drill-down
- * Coverage, LTFU, and 1-3-6 compliance trends over time
- * Identification of underperforming hospitals for targeted support
- * Data export for national UNHS programme evaluation and policy reporting

The national dashboard does not display individual baby records at any level -- only aggregate programme metrics. This is a hard architectural constraint, not a configuration option.

5. Alert System -- Transfer-Out, Missed Follow-Ups, Reminders

DengarTrack operates an automated, rules-based alert engine that removes the burden of manual follow-up tracking from coordinators. Every refer case is tracked from the moment it is recorded until a confirmed outcome is entered.

5A Automated LTFU Risk Alert Timeline

Once a REFER result is entered, the system automatically initiates the following timeline without any coordinator action:

Trigger	When Channel Action
Reminder 1	Day 7 after REFER result WhatsApp / SMS to caregiver Appointment reminder in Bahasa Melayu and English
Reminder 2	Day 14 -- no response WhatsApp / SMS to caregiver Second reminder with hospital contact number

Amber Escalation

Day 28 -- no outcome recorded | In-app alert to Coordinator | Case flagged amber in follow-up queue for manual follow-up

Red Escalation

Day 60 -- no outcome recorded | In-app alert to Coordinator + Hospital Admin | Case marked high-risk LTFU; visible in dashboard LTFU count

All SMS and WhatsApp message templates are pre-written in Bahasa Melayu (with English toggle), editable by the coordinator. Messages contain the hospital name and contact number only -- no clinical details are included in external messages.

5B Transfer-Out Case Alerts

When a baby is referred to another hospital for diagnostic ABR or intervention, the sending coordinator initiates a secure in-app handover. The alert system tracks the handover lifecycle:

- * Sending coordinator receives confirmation when the receiving coordinator's system acknowledges the handover
- * If no response from receiving hospital within 72 hours -- escalation alert sent to both coordinators
- * Sending hospital dashboard shows 'Handover Pending' status until receiving hospital accepts
- * Outcome notification sent back to sending coordinator when receiving hospital records a result
- * No baby ever falls through the inter-hospital gap silently -- the system maintains accountability on both sides

No baby name, mother IC number, date of birth, or clinical identifier is transmitted via WhatsApp or any external messaging platform at any point in this flow. Baby System ID only.

5C WhatsApp Integration for Parent Communication

DengarTrack integrates with the WhatsApp Business API -- not consumer WhatsApp -- to send automated, compliant notifications to parents and caregivers:

- * Automated appointment reminders at Day 7 and Day 14 after a REFER result
- * Appointment confirmation messages once a follow-up is booked
- * General programme information messages (configurable by coordinator)
- * SMS fallback if WhatsApp is unavailable or the number is not WhatsApp-enabled

The critical distinction: messages to parents go OUT from DengarTrack via WhatsApp Business API. Clinical data never goes INTO WhatsApp from hospital staff. This eliminates the PDPA violation while preserving the reach of WhatsApp as a communication channel.

Future integration with MySejahtera's notification system (for parents who have their child registered as a dependent) is a planned Phase 3 capability, enabling UNHS follow-up reminders to appear directly within the government health app parents already use.

6. Integration with Other Healthcare Data Systems

DengarTrack is built on HL7 FHIR R4 -- the international standard for healthcare data exchange. This is not a future aspiration; it is a present architectural decision that makes every integration listed below technically feasible without rebuilding the core platform.

6A Current and Near-Term Integrations (Phases 1-3)

- * i-Jejak / Hospital Birth Registration System -- receive Baby System ID at the point of delivery, eliminating manual ID entry by the screener
- * KKM National UNHS Registry -- push screening data directly to the national registry, replacing the current manual Excel submission process
- * HIS/iPerak (Hospital Information System) -- pull patient demographic data to pre-populate screening records, reducing data entry time further
- * JPN National Registration Department -- verify baby identity via IC number linkage for inter-hospital handover accuracy
- * WhatsApp Business API (Twilio) -- automated parent communication as described in Section 5

6B Medium-Term Integrations (Phase 3-4)

- * MySejahtera -- push UNHS REFER status as a health notification visible to parents in their MySejahtera account; supports the MOH '1 Person, 1 Record' vision
- * Klinik Kesihatan HIS -- flag unresolved REFER cases in primary care records so family physicians and community nurses can prompt parents at immunisation and postnatal visits
- * Diagnostic Audiology Clinic Systems -- receive diagnostic ABR outcome to automatically close the 1-3-6 benchmark tracker loop in DengarTrack
- * Early Intervention Programme (EIP) Systems -- receive intervention enrolment confirmation as the final outcome in the UNHS care pathway

ASEAN and LMIC Expansion Architecture

The modular architecture is designed for replication across ASEAN public health systems facing equivalent UNHS sustainability challenges. Each module can be activated independently:

- * Module activation configurable per country -- a system with no national registry can run M1 to M4 without M6
- * Language configuration -- Bahasa Melayu template can be replaced with any language without code changes
- * Device compatibility at Android 9+ covers the majority of devices in use across ASEAN LMIC health systems
- * FHIR R4 compliance ensures interoperability with any national health information system that adopts international standards

7. Data Protection Throughout All Phases

Data protection is not a feature added to DengarTrack -- it is a design constraint applied from the first line of code. Every architecture decision is evaluated against the Personal Data Protection Act 2010 (PDPA) before implementation.

7A Encryption -- Data in Transit and at Rest

- * TLS 1.3 -- all data transmitted between the mobile app and server is encrypted using Transport Layer Security 1.3, the current banking and financial services standard
- * AES-256 -- all data stored in the cloud database is encrypted at rest using 256-bit Advanced Encryption Standard
- * SQLite encryption -- offline data stored on the device is encrypted; a lost or stolen tablet cannot expose patient data
- * AWS / Azure Malaysia Region -- all data hosted within Malaysia's geographic borders, satisfying PDPA data residency requirements

7B Role-Based Access Control (RBAC)

Each user sees only what their role requires. These restrictions are enforced at the API level -- they cannot be bypassed by modifying the app interface:

Role	What they can access
Screener	Own screening entries for current shift. Cannot see other wards, coordinator data, or reports.
Coordinator	All data for their hospital -- dashboard, follow-up queue, handover, reports. Cannot see other hospitals' baby records.
Hospital Admin	Hospital-level summary metrics and audit logs only. No individual baby records.
KKM / MoH	Aggregate national dashboard and report download only. No individual hospital or baby data.

7C Anonymisation and Minimum Necessary Data

- * Baby System ID only -- no baby name or mother IC number is ever transmitted between hospitals. Raw identifiers stay within the originating hospital's system.
- * Parent communication messages contain no clinical details -- only the hospital name, contact number, and a generic reference to a follow-up appointment
- * KKM national dashboard displays aggregate counts only -- no individual baby or hospital worker records are visible at the national level
- * Screener shift summary masks baby IDs to the last 4 digits -- '*****1234' -- for privacy even within the ward

7D Immutable Audit Trail

- * Every action in DengarTrack -- screening entry, status update, handover initiation, report export -- is logged with a timestamp, user ID, and device identifier
- * Audit logs are append-only: no user, including system administrators, can edit or delete an audit record
- * Handover records are immutable after creation -- neither sending nor receiving coordinator can alter the record, satisfying PDPA's accountability requirement
- * Audit trail is exportable as a PDF for hospital JCI or MS Quality accreditation and PDPA compliance evidence
- * Log retention: minimum 7 years, consistent with Malaysian medical records legislation

7E External Security Validation

- * External cybersecurity penetration test required before any hospital pilot deployment
- * PDPA Data Protection Officer consultation completed before pilot
- * Ethics approval for each pilot phase obtained via NMRR -- Amendment No. 1 already approved
- * Ongoing vulnerability scanning post-deployment via cloud provider security tools
- * Incident response protocol documented -- data breach notification procedure in place before go-live

Implementation Pathway

DengarTrack does not require a big-bang national rollout. It is designed to prove value at small scale before expanding:

Phase	Timeline Scope Key milestone
Phase 1 -- Validate	Apr-Jun 2026 Design + Figma prototype Coordinator validation in 2-3 hospitals; i-ACHIEVE 2026 presentation 16 Jun
Phase 2 -- Build MVP	Jul-Dec 2026 M1 + M2 + M3 functional Working Android app ready for Pilot 1
Phase 3 -- Pilot 1	Oct-Dec 2026 2-3 hospitals Usability study (SUS target ≥ 68); system refinement
Phase 4 -- Pilot 2	Jan-Jun 2027 6-8 hospitals Effectiveness study; primary outcome: ≥ 15 pp LTFU reduction
Phase 5 -- Scale	2027-2028 All 48 KKM UNHS hospitals National rollout; MySejahtera + Klinik Kesihatan integration

DengarTrack in One Paragraph

DengarTrack is a mobile-first digital platform that enables ward screeners to record OAE and AABR results in under 60 seconds at the cot side, automatically tracks every referred baby through the 1-3-6 pathway, alerts coordinators and parents when follow-ups are missed, generates KKM monthly reports with one click, and replaces the illegal WhatsApp clinical data sharing practice with an encrypted in-app system -- while remaining fully compliant with the Personal Data Protection Act 2010 and designed to integrate with MySejahtera, i-Jejak, and KKM national health information systems. Built on published evidence from 19 Malaysian public hospitals. Targeting a reduction in LTFU rate from 29% to $\leq 10\%$.

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